

1. General Comparison (22 Criteria)

Criterion	Rule-Based Engine	Fine-Tuned Model
Speed	<30ms	200-800ms on CPU
Infrastructure	None required	Needs GPU
Cost	Free	Training + Hosting cost
Explainability	Full — reasons per rule	Black box
Modifiability	Add rule in minutes	Requires full retraining
Dev Time	~1 week	Weeks + experiments
Spam Detection	Moderate	Excellent
Phishing Detection	Excellent (20 rules)	Moderate
BEC Detection	Good	Moderate
Link Mismatch	Excellent	Weak
Homoglyph / Unicode	Excellent	Moderate
SPF / DKIM / DMARC	Excellent	Weak
Reply-To Mismatch	Excellent	Weak
Received Headers	Excellent	Weak
First Contact Risk	Good (DB-based)	Not available
Unknown Threats	Needs new rule	Can detect patterns
Generalization	Rule-dependent	Learns patterns
Arabic / Regional	Add keywords easily	Needs Arabic dataset
Gmail Users	Complements Gmail	Redundant with Gmail
Outlook Users	Full protection	Limited add-on
Custom IMAP Users	Only protection layer	Blind to local context
Graduation Project Fit	Full — explainable to committee	Hard to prove results

2. Detection Rules Strength (All 20 Rules Implemented)

#	Rule	Our Engine	Gmail	Fine-Tuned
1	Spam Keywords (38+)	Moderate	Excellent	Excellent
2	Fuzzy Keyword Match	Good	Moderate	Excellent
3	Temp Email Domains	Good	Good	Moderate
4	Excessive CAPS / Exclamation	Moderate	Moderate	Moderate
5	Excessive Links	Moderate	Moderate	Moderate
6	Typosquatting (Levenshtein)	Good	Moderate	Excellent
7	Brand / Domain Mismatch	Good	Moderate	Excellent
8	Urgent Language Patterns	Moderate	Moderate	Good
9	IP-based URLs	Good	Good	Good
10	Shortened URLs	Moderate	Good	Good
11	Suspicious TLDs	Moderate	Moderate	Good
12	Link Mismatch (visible vs actual)	Excellent	Weak	Weak
13	First Contact Risk Scoring	Good	Weak	Weak
14	BEC Language Detection	Good	Weak	Good
15	Attachment Risk Context	Good	Weak	Moderate
16	Brand Abuse Detection	Good	Weak	Good
17	Homoglyph / Unicode Spoofing	Excellent	Weak	Moderate
18	Reply-To Mismatch	Excellent	Weak	Weak
19	Received Headers Chain	Excellent	Weak	Weak
20	SPF / DKIM / DMARC Validation	Excellent	Moderate	Weak

3. Provider Context

Provider	External Filter	Rule Engine Value	Fine-Tuning Value
Gmail	Excellent	Complements Gmail on advanced phishing	Redundant — Gmail already handles spam
Outlook	Moderate	Full protection on advanced rules	Limited add-on over Outlook
Custom IMAP	None	Only protection layer — no alternative	Blind to local/custom context

4. Unique Advantages of Rule-Based Engine

Capability	Gmail	Fine-Tuned Model
Link Mismatch Detection	Not available	Not available
First Contact Risk Scoring	Not available	Not available
Received Headers Chain Analysis	Not available	Not available
Reply-To Mismatch Detection	Partial only	Not available
Homoglyph / Unicode Spoofing	Very weak	Moderate
SPF/DKIM/DMARC as Risk Score	Displays only	Not available
Full Explainability (reasons[])	Black box	Black box

Final Verdict: Rule-Based Engine (20 rules) + Gmail/Outlook Filtering + AI Agent = Optimal Architecture